

THATO BARUTI

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ML · Optimisation and Uncertainty · Simulation Pipelines · Engineering AI Platforms

Applied

PROFILE

Applied ML engineer completing an MSc in Computational Fluid Dynamics at Cranfield University, building deployment-facing AI systems for simulation-driven engineering and CAE workflows. My work spans dataset and graph construction, model training, evaluation gates, uncertainty calibration, optimisation workflows, high-fidelity CFD validation, and API/UI layers that expose models safely. I am strongest at turning research ideas into auditable engineering tools with clear contracts, traceability, and realistic limitations rather than treating model accuracy in isolation as the finish line.

SELECTED ENGINEERING EXPERIENCE

AeroGenic Applied ML and Platform Stack

MSc Research, Cranfield University

2025 – Ongoing

- Built the AeroGenic codebase as an end-to-end engineering AI stack for aerodynamic coefficient prediction, combining graph neural surrogates, baselines, calibration tools, evaluation gates, report generation, and deployment-facing model-selection logic.
- Added FastAPI endpoints and a Gradio workbench that convert uploaded geometry and flight requirements into traceable wing-design outputs, confidence scores, exported geometry, source-case provenance, and machine-readable JSON reports; best held-out models reached 3.18% for C_L , 2.60% for C_D , and 3.83% for C_{My} , with 2.49% on C_D for the polar model.

Simulation, Optimisation, and Selection Workflows

Cranfield University

2025 – Ongoing

- Implemented requirement-conditioned Pareto optimisation across drag, trim, lift overshoot, span, and training-manifold risk; generated 63,393 candidates, retained 15,031 feasible lift-band designs, and extracted 1,012 non-dominated solutions.
- Built candidate-selection logic for higher-fidelity CFD follow-up using residual size, uncertainty width, aerodynamic stress variables, and representative anchor-case coverage so every recommended geometry carries a traceable validation status into the next gate.

HPC CFD Commissioning and Validation Pipeline

Cranfield University HPC Infrastructure

2025 – Ongoing

- Automated OpenFOAM case generation and debugging across meshing, decomposition, solver stability, module-path propagation, and function-object compatibility issues; encoded the structural fixes directly into the generator so regenerated cases inherit them automatically.
- Managed a 32-case RANS validation campaign with 29 returned cases and 26 strict-pass evidence cases, then used the CFD layer to build grouped geometry-held-out residual correction reducing corrected MAE to 0.0677 for C_L , 0.0104 for C_D , and 0.401 for C_{My} .

EDUCATION

MSc Computational Fluid Dynamics

Cranfield University, UK

Oct 2025 – Aug 2026 (Expected)

On track for distinction; selected modules include Data Analysis & Post-Processing (87%), Introduction to Fluid Mechanics & Heat Transfer (87%), Turbulence Modelling, Numerical Methods & HPC, Grid Generation, and Numerical Modelling.

BEng Aerospace Engineering

University of Sheffield, UK

Jun 2025

Awards: University of Sheffield International Undergraduate Scholarship; Botswana Government Top Achievers Scholarship.

TECHNICAL SKILLS

Languages and ML	Python; C++; PyTorch; PyTorch Geometric; NumPy; scikit-learn; FastAPI; Gradio
ML Engineering	Graph neural networks; surrogate modelling; calibration; reliability ranking; residual correction; optimisation pipelines; model comparison gates
Platform	REST-style APIs; JSON reporting; workflow traceability; figure and report generation; evaluation harness design
Simulation	OpenFOAM; ANSYS Fluent; UCNS3D; Linux HPC workflows; PBS; SLURM; MPI-parallel job pipelines; ParaView; Tecplot
Domain	Aerodynamics; wing design; CFD validation; simulation-driven product and CAE workflows